System Architecture & Design Specification

Project: Premium Fingerprint Security System

Platform: ATmega32 Microcontroller

OS: FreeRTOS

1. Hardware Architecture & Interfaces

The system orchestrates multiple hardware modules to ensure a secure access control environment.

- **Main MCU:** ATmega32 operating at a defined clock frequency.
- **Biometric Sensor:** R307 Optical Fingerprint Module interfacing via **UART** (Baud Rate: 57600).
- **Non-Volatile Storage:** 24C16 External EEPROM interfacing via **I2C**.
- **User Interface:** 16x2 LCD and 4x4 Matrix Keypad interfacing via **GPIO**.
- **Security Monitoring:** LDR Sensor interfacing via **ADC** for enclosure breach detection.
- **Actuators:** DC/Servo Motor for locking mechanism and Piezo Buzzer for alarms.

2. Database Schema & Memory Management

To optimize the limited RAM (2KB) of the ATmega32, the system utilizes a direct mapping architecture (Array of Structs) instead of dynamic nodes.

- **Data Structure:** UserRecord_t containing UserID (2 bytes) and Privilege (1 byte).
- **Memory Footprint:** Maximum capacity is strictly bounded to **100 users**, consuming exactly 300 bytes of RAM.
- **Boot Initialization:** On system boot, the initialization function performs a bulk read from the **EEPROM** to populate the RAM array, minimizing I2C bus traffic during standard operation.

- **Search Algorithm:** The system utilizes **Linear Search** within the RAM array for authentication, providing O(N) time complexity which executes in sub-microseconds for N=100.
- **ID Synchronization:** The unique ID generated by the R307 sensor acts as the primary key (UserID) across both the sensor's internal flash and the system's EEPROM.

3. RTOS Architecture & Concurrency

The system relies on a preemptive FreeRTOS scheduler to manage asynchronous hardware events and maintain system responsiveness.

Task Name	Priority	Core Functionality
Task_SecurityMonitor	High (3)	Samples LDR sensor continuously. Triggers lockdown and alarm upon detecting enclosure tampering.
Task_Fingerprint	Medium (2)	Polls UART for incoming R307 data packets, executes authentication logic, and drives the lock mechanism.
Task_UI_Manager	Low (1)	Manages Keypad inputs and LCD updates for the Admin Mode (Add/Delete/Search).

- **Resource Protection:** A strictly managed **Mutex** (Database_Mutex) protects the EEPROM and RAM Array. Any task requiring read/write access must acquire this Mutex to prevent data corruption during simultaneous user authentication and admin modifications.

4. System Integrity & Fallback Mechanisms

To achieve industrial-grade reliability, the architecture incorporates strict fault-tolerance protocols.

- **Watchdog Timer (WDT):** Configured to reset the MCU automatically if any RTOS task enters a deadlock state or the I2C bus hangs indefinitely.

- **Transaction Rollback:** Adding or deleting a user requires dual-memory modification (R307 internal flash + System EEPROM). The system verifies the success of the R307 operation before executing the EEPROM transaction.
- **Data Mismatch Flagging:** If a deletion succeeds on the R307 but fails on the EEPROM, the system raises a critical error flag on the UI, blocking further modifications until a manual admin override restores database synchronization.